# AB-U2T-A030B-SXX
# (X=blank, A-Z)
# User Manual



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# Preface

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The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

This document gives a description of the charging module, including the features, performance, appearance, structure, working principles, installation, operation and maintenance etc. Please save the manual after reading, in order to consult in the future.



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## Revision history

This section records the date and change list of the document when updated.

| Version | Revise information | Author | Date |
|---|---|---|---|
| S00 | First Draft | Jiahao.Liu | 2023-11-04 |
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# Safety Precautions

## 1 Statement

The Company shall not be liable for any of the following situations:

- Use the equipment in harsh environments beyond those described in this manual;
- Use the equipment in any installation and practical environment outside the requirements of the relevant international standards;
- Change the product or modifies the software code without authorization;
- Operate without accordance to the operation instructions and safety warnings in the product and documentation;
- Equipment damaged caused by unnatural environment.

## 2 General safety precautions

### 2.1 Electrical safety

- The electrical connection must be carried out in strict accordance with the description in the manual and electrical wiring schematic diagram;
- Before power on, please confirm that the equipment has been properly grounded and check that the connection of wiring is correct;
- Special tools shall be used for relevant electrical operations;
- When the equipment needs to be moved or rewired, the power supply shall be disconnected and the corresponding operation shall be carried out after the equipment is completely powered off.

### 2.2 Personnel safety

- Personnel engaged in various electrical operations and equipment installation must hold relevant operation qualifications;



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- During the installation, maintenance and other operations of the equipment, relevant personnel shall take appropriate protective measures as required, such as wearing anti-static work clothes, anti-static gloves, etc., and remove conductive objects such as jewelry and watches to avoid electric shock or burns.

## 2.3 Operations safety

- Please read this manual carefully and pay attention to the safety rules and correct operations;
- Please operate as required;
- Avoid using the equipment in direct sunlight, rainy or humidity environment;
- Do not put the equipment in the fire or near the electric heater, hot stove and other similar hot source;
- In case of fire, please correctly use dry powder fire extinguisher to extinguish the fire. If you use liquid fire extinguisher, there will be danger of electric shock.



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# Chapter 1 Introduction of charging module

The charging module named AB-U2T-A030B-SXX can be used in constant current or constant voltage mode with 3P3W+PE inputs without neural line. This model has the advantages of high efficiency, high power factor, high-power density, low current harmonics. Especially, optimal design for low electromagnetic radiation and interference, low noise, high reliability. This product is designed with glue protection(top layer potting), which is suitable for harsh environment such as high humidity and heavy dust.

**Key Features & Benefits**

- Output power up to 30kW
- Typical efficiency 97.2% (half load to full load @ 400Vac IP,800Vdc OP)
- AC input range: 260 – 530 Vac (Line to line voltage)
- DC output voltage range 150 - 1000 VDC, constant power output at 300 - 1000 VDC
- OTP, OVP, UVP, OCP protection
- Operation temperature -40°C to 55°C at full load
- CAN bus communication interface

## 1.1 Model name

| Model | description |
|---|---|
| AB-U2T-A030B-SXX | 30kW isolated charging module (three phase) |



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## 1.2 Structure

### 1.2.1 Outward appearance

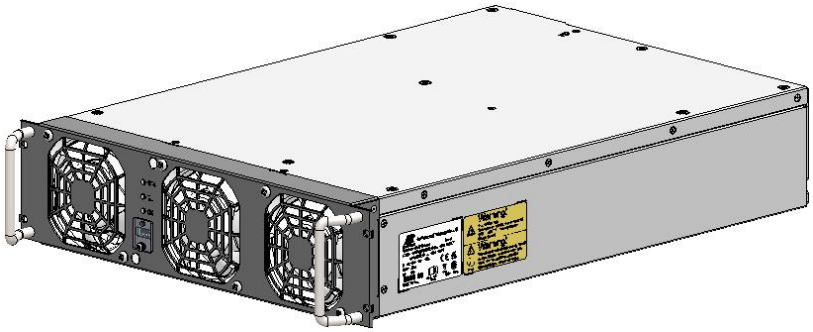


Figure 1-1 Outward appearance

### 1.2.2 Front panel

There are indicators and address setting board on the front panel of the charging module as shown in Figure 1-2 and 1-3. The instruction table description is shown in Table 1-1.

Table1-1 Indicator light description

| | Situation | Green (OK) | Yellow (Derating) | Red (RFA) | status |
|---|---|---|---|---|---|
| **LED indicators** | 1 | Blink | ON | OFF | Module is in initialization state |
| | 2 | Blink | OFF | OFF | Module is standby, waiting for the command from system |
| | 3 | ON | OFF | OFF | Module is operating correctly |
| | 4 | ON | ON | OFF | 1. Input AC voltage derating<br>2. Output DC Voltage derating<br>3. High temperature power derating |
| | 5 | OFF | OFF | Blink | Alarm happened, such as OTP shutdown, OVP/UVP shutdown, Internal or External communication fail and so on |
| | 6 | OFF | OFF | ON | Module failure |
| | 7 | OFF | OFF | OFF | Aux power is off or fail |



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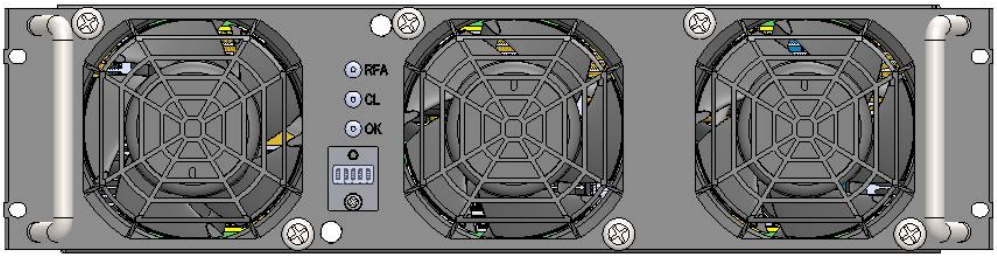



Figure1-2 Front panel

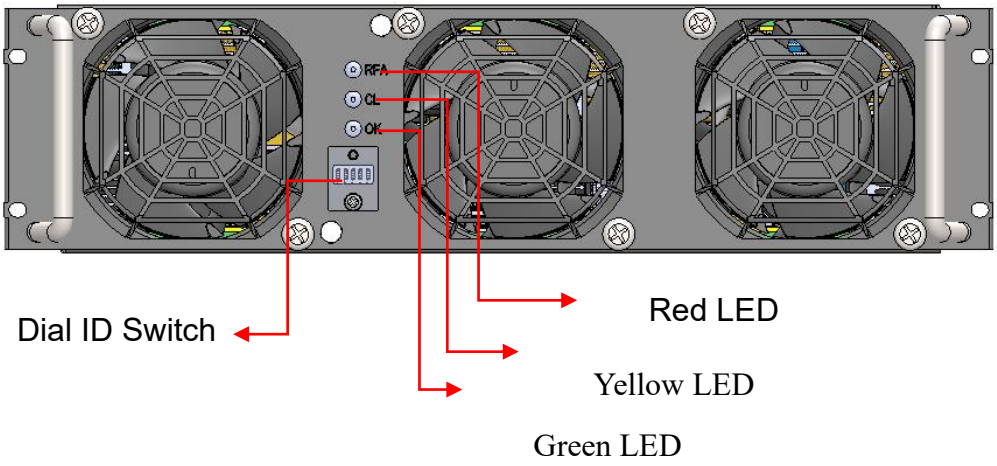
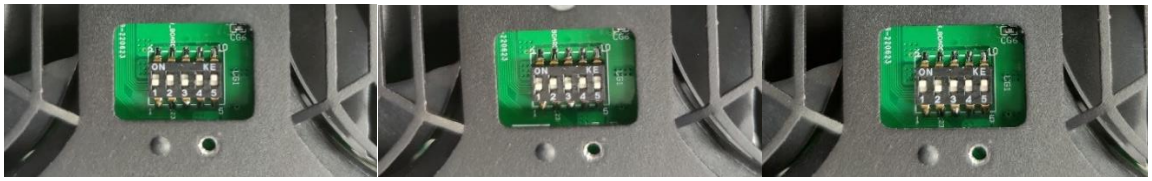



Figure1-3 Details on the front pane

Note: The ID is binary 8421 code, the right side is high bit, and the left side is low bit; The dial down indicates low, indicating 0, and the dial up indicates high, indicating 1. Refer to figure 1-4 for examples. Refer to the conversion information in the following table for specific corresponding information. The maximum effective address for dialing is 31.



Address: 26(11010)　　Address: 10(01010)　　Address: 29(11101)

| Address | Switch | Address | Switch | Address | Switch | Address | Switch | Address | Switch | Address | Switch |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 00000 | 6 | 00110 | 12 | 01100 | 18 | 10010 | 24 | 11000 | 30 | 11110 |
| 1 | 00001 | 7 | 00111 | 13 | 01101 | 19 | 10011 | 25 | 11001 | 31 | 11111 |
| 2 | 00010 | 8 | 01000 | 14 | 01110 | 20 | 10100 | 26 | 11010 | | |
| 3 | 00011 | 9 | 01001 | 15 | 01111 | 21 | 10101 | 27 | 11011 | | |
| 4 | 00100 | 10 | 01010 | 16 | 10000 | 22 | 10110 | 28 | 11100 | | |
| 5 | 00101 | 11 | 01011 | 17 | 10001 | 23 | 10111 | 29 | 11101 | | |

Figure1-4 Examples of dial ID switch



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### 1.2.3 Module interface definition

The back end of the charging module has an AC interface and a DC interface shown in Figure 1-5.

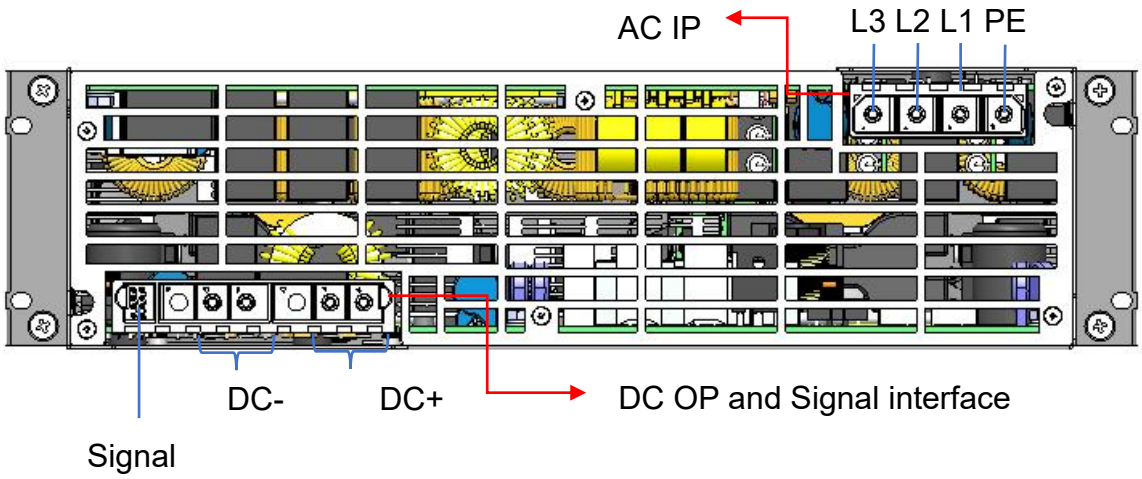



Figure1-5 Schematic diagram of module back-end input and output port definition

### 1.2.4 Module mechanism size

The module and package size are shown in Figure 1-6 and 1-7.

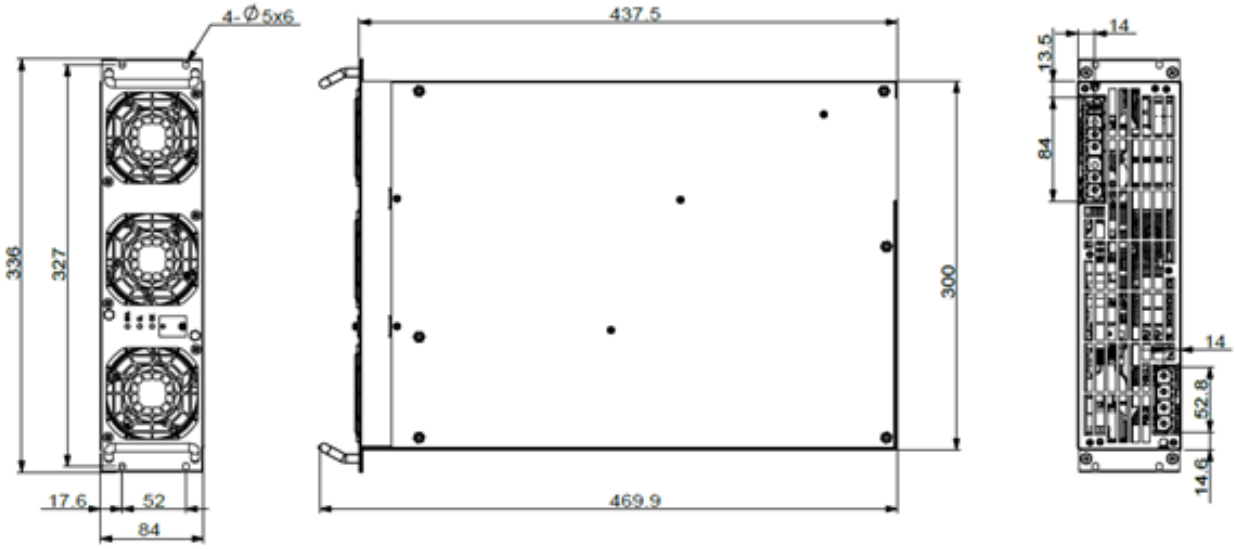



Figure 1-6 Module Structure Size (Unit: mm)



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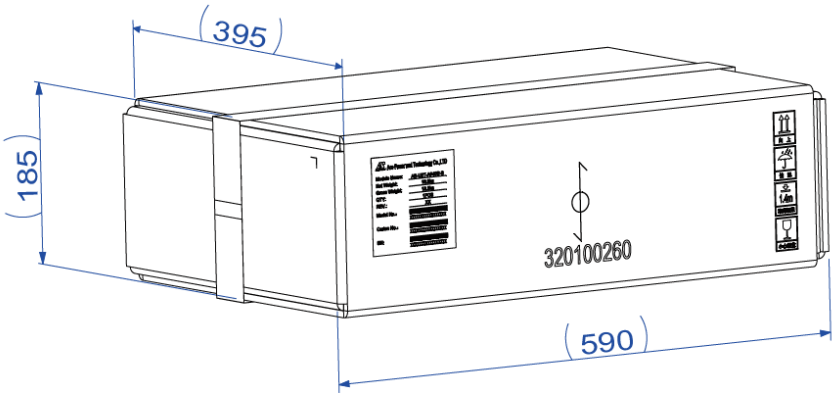



Figure 1-7 Packing Specifications (Unit: mm)

A nameplate label is affixed to the charging module, as shown in Figure 1-8.




Figure1-8 Nameplate label

## 1.3 Technical Parameter

### 1.3.1 Main characteristics

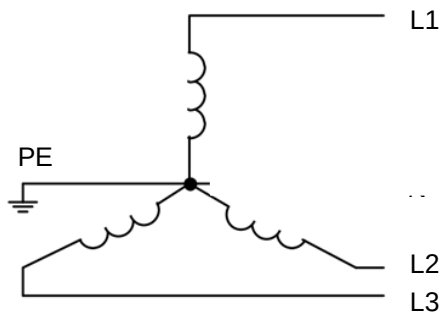
#### 1.3.1.1 Configuration of AC power grid

The charging module is able to operate at power grid configuration that shown as below:

| Type | Configuration |
|---|---|



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| | |
|---|---|
| three-phase/3-wire+PE<br>(L1, L2, L3, PE) |  |

### 1.3.1.2 Spec.

| Parameter | Description/condition | Min | Nor | max | Unit |
|---|---|---|---|---|---|
| AC input side parameters | | | | | |
| Input voltage three phase | Line to line voltage | 260 | 400 | 530 | Vac |
| Input current | @ input L-L 320Vac full load | | | 60 | A |
| Line frequency | | 47 | 50/60 | 63 | Hz |
| Leakage current | @Vin=400Vac | | | 3.5 | mA |
| Power factor | Full load | 0.99 | | | |
| THD | Full load | | | 5% | |
| Efficiency | Max. Value@ 800Vdc half load | | 97.4 | | % |
| | Max. vaule@ 800Vdc full load | | 97.2% | | % |
| DC output side parameters | | | | | |
| Output voltage | | 150 | 1000 | 1000 | Vdc |
| Output current | Max. value, @Vout=300V and Pout=30KW | | | 100 | A |
| Output power | Vin=320~530Vac, Vout=300~1000Vdc | | 30 | 30 | KW |
| Startup current | % of current at normal operation | | | 110 | % |

### 1.3.1.3 Measurement accuracy and setting definition

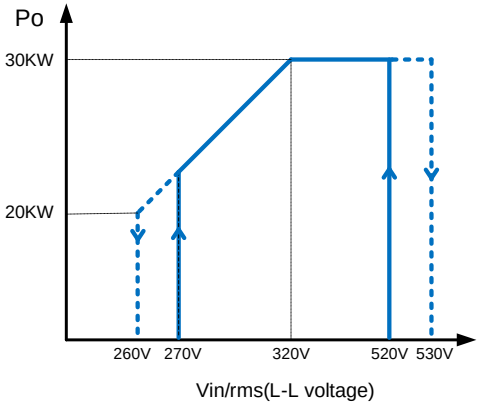
| Parameter | Description/condition | Min | Nor | max | Unit |
|---|---|---|---|---|---|
| AC side parameters | | | | | |
| AC voltage accuracy | Measurement @ Nominal voltage 400Vac | | | +/-2V | Vac |
| | | | | | |
| DC output side parameters | | | | | |

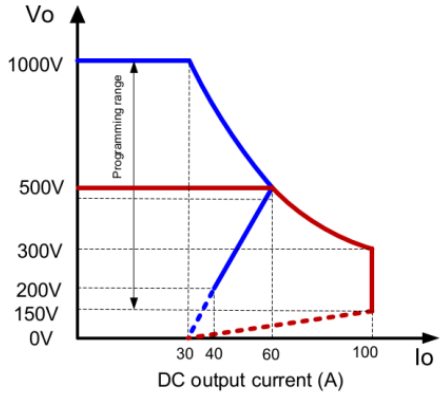


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| | | | | | |
|---|---|---|---|---|---|
| DC output voltage | measurement accuracy @ 150~1000Vdc range | | | +/-0.5% | |
| | Setting definition | 0.5 | | | V |
| DC output current | Measurement accuracy | | | 1 | A |
| | Current setting definition | 0.1 | | | A |
| Ambient temperature | Measurement accuracy | | | +/- 3 | °C |

### 1.3.1.4 Output power de-rating curve




### 1.3.1.5 Protections

| Parameter | Description/condition | Min | Nor | max | Unit |
|---|---|---|---|---|---|
| AC OVP | >530V(±4Vrms) | | 530 | | Vac |
| AC UVP | <260V(±4Vrms) | | 260 | | Vac |
| AC OFP | >63Hz(±0.2Hz) | | 63 | | Hz |
| AC UFP | >63Hz(±0.2Hz) | | 47 | | Hz |
| DC OVP | Software OVP(High Voltage Mode) | | Vcmd*1.1 | | Vdc |
| DC OVP | Software OVP(Low Voltage Mode) | | Vcmd+50V | | Vdc |
| DC UVP | Hardware OVP | | Vcmd+100V | | Vdc |
| OTP | Converter shut down at T_amb higher than | | 75 | | °C |

## 1.3.2 Monitoring and control signals

| Parameter | Description/condition |
|---|---|
| CAN (CAN1H, CAN1L) | 125 Kbit/s, this CAN bus is design for communicate with system controller to operate the module |
| EN-sys, CGND-sys | Enable signal（operation forbidden when in short circuit state） |



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### 1.3.3 Environmental Specifications

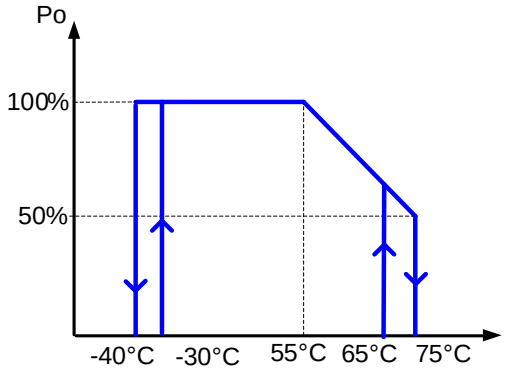
| Parameter | Description/condition | Min | Nor | max | Unit |
|---|---|---|---|---|---|
| Operation altitude | | | | 4000 | m |
| Operation Temperature | | -40 | | 75 | °C |
| Storage temperature | | -40 | | 85 | °C |
| Relative humidity | | 0 | | 95% | |
| Protection | glue protection(top layer potting) is used to reduce the failure rate caused by dust accumulation and condensation | | IP20 | | |
| Pollution degree | IEC61851-1/ UL2202 | | II/III | | |
| Acoustic noise | 400V/800Vdc Full Load@25℃ | | 51 | | dB |
| Overvoltage degree | IEC61851-1/ UL2202 | | 2 | | |

### 1.3.4 Cooling Specifications

| Parameter | Description/condition | Min | Nor | max | Unit |
|---|---|---|---|---|---|
| Cooling method | Forced air-cooling | | | | |
| Maximum ambient temperature | +75°C (55~75°C linear derating to 50% of full load) | -40 | | 75 | |

### 1.3.5 Thermal derating curve

The module can operate in the ambient temperature up to 75°C with derating over 55°C. The derating curve is shown below with 50% of rated power at 75°C. Power derating occurs when the temperature of the internal parts of the module is too high under some extreme conditions




### 1.3.6 Safety, Regulatory and EMI Specification

| Parameter | Description/condition | Criterion |
|---|---|---|
| Emission requirements: IEC 61851-21-2 | | |



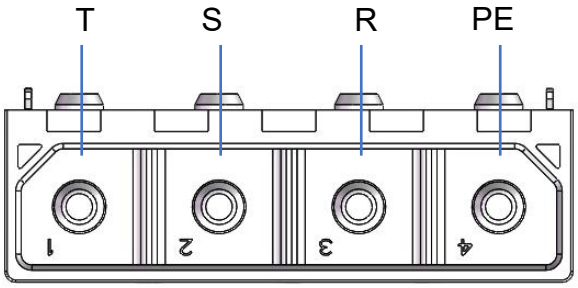
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| | | |
|---|---|---|
| Radiated Emission | CISPR16-2-3 | Class B |
| Conducted Emission | CISPR 16-2-1 | Class B |
| Harmonics input current | EN 61000-3-12 | |
| Immunity requirements: IEC 61851-21-2 | | |
| Electrostatic Discharge | IEC61000-4-2 | |
| Radiated RF field | IEC61000-4-3 | |
| Electrical Fast Transient (EFT)/burst | IEC61000-4-4 | |
| Surge immunity | IEC61000-4-5 | |
| conducted RF field | IEC 61000-4-8 | |
| Hi-pot test | AC input to HV output:<br>AC input to chassis<br>HV output to chassis: | Refer to IEC 61851 |
| Insulation resistance | The insulation resistance between each independent circuit and the ground (chassis)<br>The insulation resistance between each circuit without electrical connection | Refer to IEC 61851 |

## 1.3.7 Input and output Connectors

### 1.3.7.1 AC side power connector

| | | |
|---|---|---|
| Manufacturer | YongHaoRun technology Co., Ltd. | |
| Manufacturer P/N | YHR MPC-4TJW1-4 | Male on the module |
| Manufacturer | YongHaoRun technology Co., Ltd. | |
| Manufacturer P/N | YHR MPC-4ZKY1 | Female on the system |



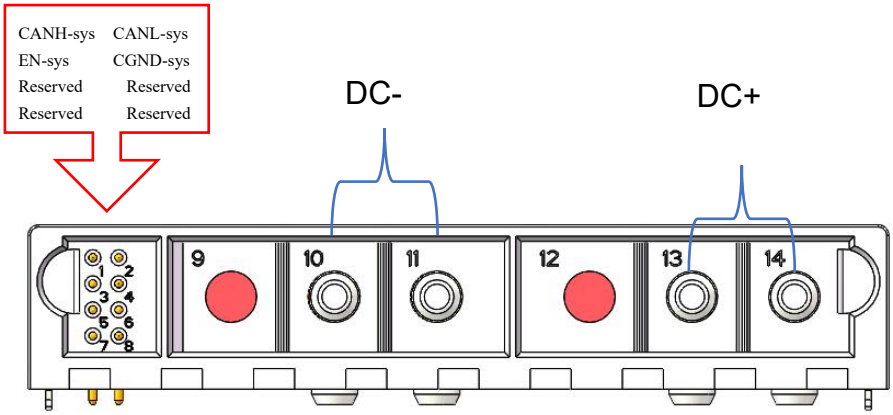

AC side connector Rear View (on-board-male)



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### 1.3.7.2 DC side power and signal connector

| | | |
|---|---|---|
| Manufacturer | YongHaoRun technology Co., Ltd. | |
| Manufacturer P/N | YHR MPC-14G12TJW1-3 | Male on the module |
| Manufacturer | YongHaoRun technology Co., Ltd. | |
| Manufacturer P/N | YHR MPC-14G12 ZKY1-2 | Female on the system |




DC side connector Rear View (on-board male)

### 1.3.7.3 DC side power and signal connector

| Pin | Signal | Description |
|---|---|---|
| AC side connector | | |
| 4 | PE | PE terminal |
| 3 | R phase | AC line 1 (Phase L1) |
| 2 | S Phase | AC line 2 (Phase L2) |
| 1 | T Phase | AC line 3 (Phase L3) |
| DC and signal connector | | |
| 13,14 | DC+ | The positive of main output |
| 11,12 | DC- | The negative return of the main output |
| 1 | CAN1H-sys | Positive terminal of CAN bus, for system control |
| 2 | CAN1L-sys | Negative terminal of CAN bus, for system control |
| 3 | EN-sys | Enable signal(operation forbidden when short to Pin4) |
| 4 | CGND-sys | Signal reference GND |
| 5 | Reserved | |
| 6 | Reserved | |



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| Pin | Signal | Description |
|---|---|---|
| 7 | Reserved | |
| 8 | Reserved | |

### 1.3.8 Mechanical Specifications

| Parameter | Description/condition |
|---|---|
| Dimensions (W*H*D) | 300mm*437.5mm*84mm |
| Weight | 17kg |
| Enclosure Materials | SGLCC |



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# Chapter 2 Installation and Application Instructions

## 2.1 Preparation before installation

### 2.1.1 Pre-installation check

- Before unpacking the outer package of the charging module, please check the outer package for visible damage, such as holes, cracks or other signs of possible internal damage, and check the model of the charging module. If there is any abnormal packaging or the charging module model is inconsistent, do not open it and contact supplier as soon as possible.
- After unpacking the charging module, please check whether the delivery is complete and whether there is any obvious external damage. If any items are missing or damaged, please contact supplier.
- Tooling Set-Up:

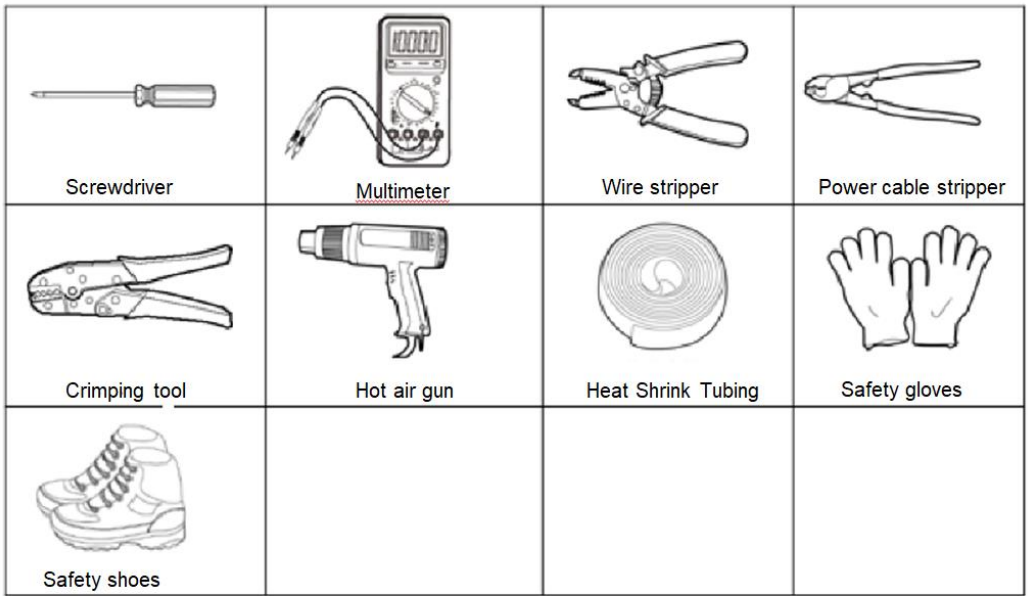



Figure 2-1 Tooling Set-Up

### 2.1.2 Prepare the cable

- The temperature resistance grade of cable insulation material shall be selected according to the ambient temperature. The temperature resistance of power cable



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insulation material shall not be less than 105 ℃, and that of communication cable insulation material shall not be less than 80 ℃.

- Users need to provide their own input connectors, output connectors and cables.

Table 2-1 Cable requirements

| Cable | Recommended Spec | Remark |
|---|---|---|
| DC cable | V≥ 1000VDC、T≥105℃、5AWG | The insulation temperature shall be less than 90℃ during operation |
| AC cable | V≥ 600VAC、T≥105℃、5AWG | |
| Signal cable | V≥ 60VDC、T≥80℃、24AWG | / |

## 2.2 Installation requirements

- The protection grade of the system shall not be lower than IP32 (indoor) or IP54 (outdoor).
- The system needs dust screen / dust cotton to prevent large particles or flocs from blocking the module air duct, and the resulting functional abnormalities or component damage are not within the scope of warranty.
- The total air volume required by a single module is ⩾ 212CFM (1m3/min=35.3CFM). If the system has N modules, the total air volume of the system shall not be less than N * 212CFM.
- The installation environment shall meet the environmental characteristics specified in the technical data. Refer to the specifications.



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# 2.3 Installation module

## 2.3.1 Installation Dimensions

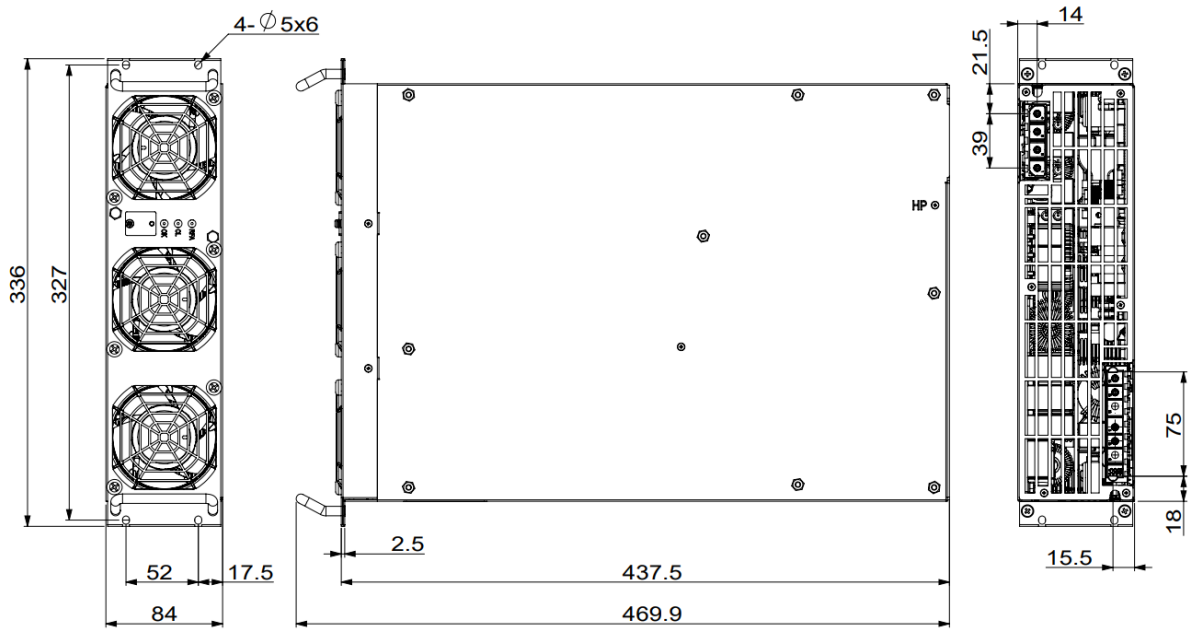



Figure 2-2 Installation Dimensions (Unit: mm)

## 2.3.2 Installation steps

Step 1 Push the module into the system slot slowly.

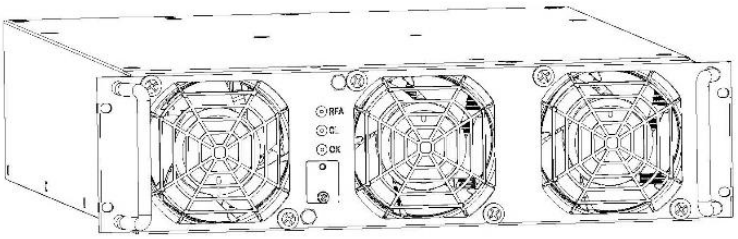


Figure 2-3 Horizontal installation



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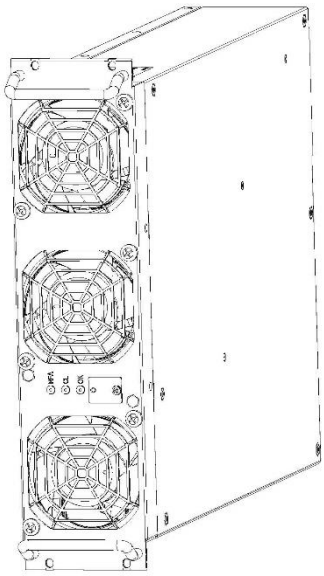


Figure 2-43 Vertical installation

Step 2 Install and tighten the screws in the middle of the panel.

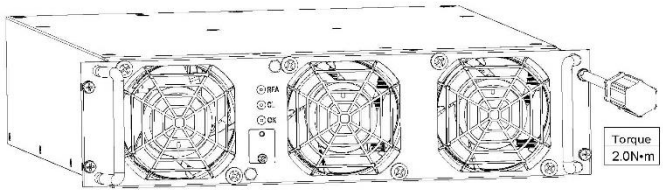



Figure 2-5 Fastening module (Horizontal)

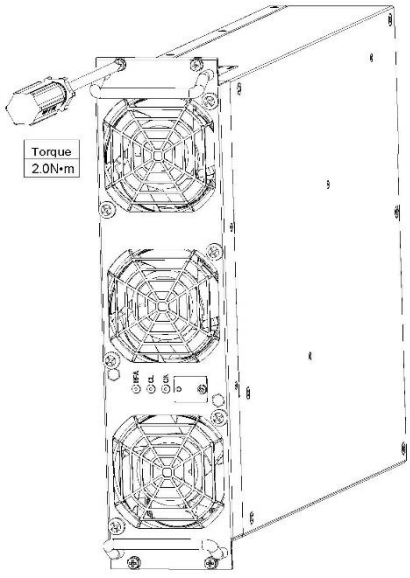



Figure 2-6 Fastening module (Vertical)

## 2.4 Accessories for test

The test cable will be supplied when the engineering prototype is provided for the first time.

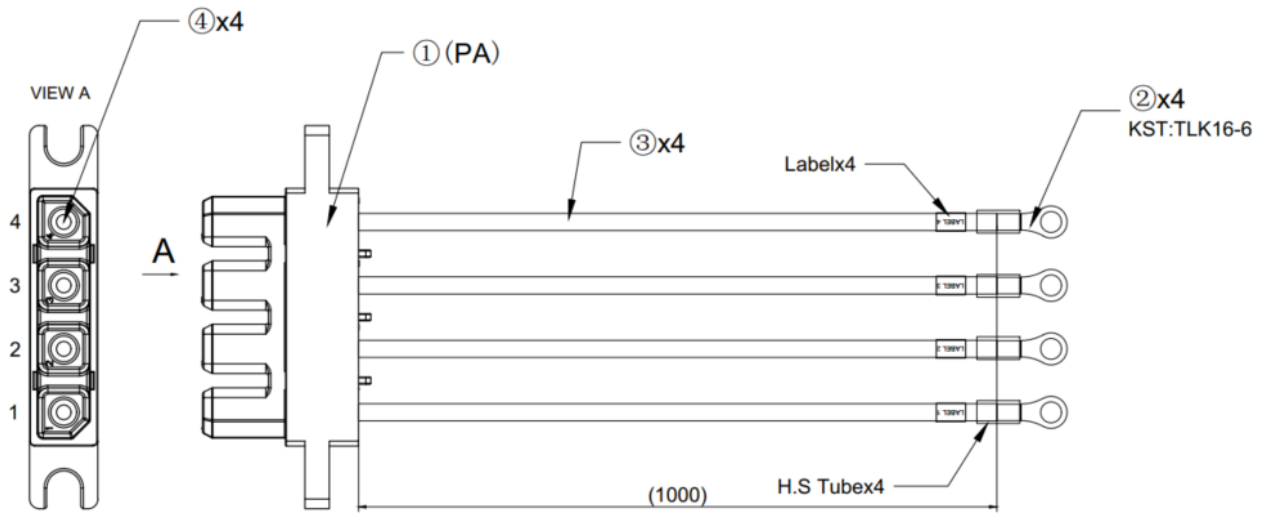


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Table 2-2 Accessories list

| Item | Function | QTY | Unit |
|---|---|---|---|
| 1 | AC test cable | 1 | PCS |
| 2 | DC and signal test cable | 1 | PCS |
| 3 | USB-CAN adapter | 1 | PCS |

### 2.4.1 AC test cable（For reference）




| LABEL | |
|---|---|
| LABEL 1 | T |
| LABEL 2 | S |
| LABEL 3 | R |
| LABEL 4 | PE |

Figure 2-7 AC test cable

Table 2-3 The connector information of AC test cables

| | | | | | |
|---|---|---|---|---|---|
| 6 | Heat shrinkable sleeve | UL，600V，125°C，VW-1; | 4 | PCS | / |
| 5 | Cable Marker | 125°C，600V，HEAT SHRINK NUMBER TUBE | 4 | PCS | / |
| 4 | Cable PE | WIRE PVC UL1015 6AWG 105°C 600V YELLOW/GREEN | 1 | PCS | / |
| 3 | Cable R,S,T | WIRE PVC UL1015 6AWG 105°C 600V BLACK | 3 | PCS | / |
| 2 | Copper terminal | YHR6.571.1068 | 4 | PCS | YHR |
| 1 | AC Input Socket | CONNECTOR, YHR MPC-4ZKY1 | 1 | PCS | YHR |
| ITEM | P/N | DESCRIPTION/VENDOR TYPE | QTY | UNIT | REMARK |



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### 2.4.2 DC and signal test cable（For reference）

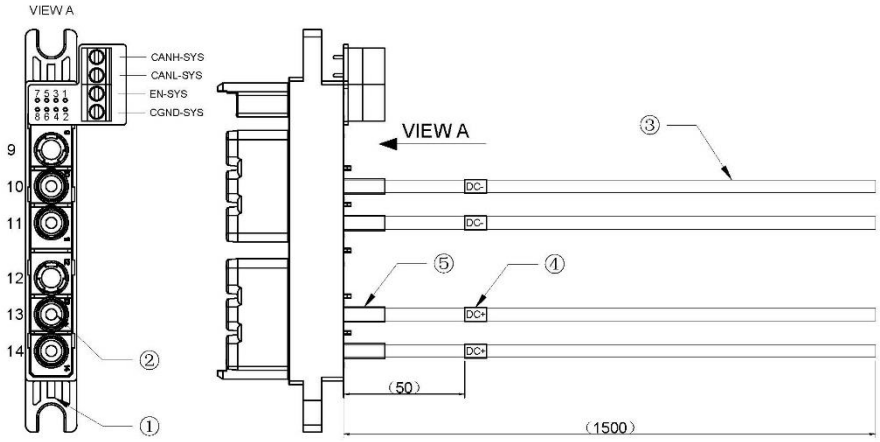



Figure 2-84 DC test cable

Table 2-4 The connector information of DC test cables

| | | | | | |
|---|---|---|---|---|---|
| 5 | Heat shrinkable sleeve | UL，600V，125°C，VW-1; | 4 | PCS | / |
| 4 | Cable Marker | 125°C，600V，HEAT SHRINK NUMBER TUBE | 4 | PCS | / |
| 3 | Cable DC | WIRE PVC UL10269 6AWG 105°C 1000V BLACK | 4 | PCS | / |
| 2 | Copper terminal | YHR6.571.1068 | 4 | PCS | YHR |
| 1 | DC Input Socket | CONNECTOR, YHR MPC-14G12 ZKY1-2 | 1 | PCS | YHR |
| ITEM | P/N | DESCRIPTION/VENDOR TYPE | QTY | UNIT | REMARK |

Notes:

1) CAN as described in the table above is used to transmit critical data communication between the system and each module, the communication bus needs to be hand-held to form a daisy-chain connection, and the terminal resistors are configured at the beginning and end of the communication bus respectively (i.e., one is configured at the system side and one terminal resistor is configured at the last terminal module).



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# Chapter 3 Maintenance Guide

## 3.1 Packaging and Transportation

When transporting the module, a separate packaging box is required, which includes the manufacturer's name, product model, manufacturing year, month, and day, and the standard number of the product. The packaging box is printed or labeled with transportation signs such as "Handle with Care", "Keep Wet", and "Up".

The packaged module can be transported by car, ship, or plane. During long-distance transportation, it should not be installed in open carriages or cabins. During transit, it should not be stored in open warehouses. During transportation, it is not allowed to be transported in the same vehicle (or other transportation means) as flammable, explosive, or corrosive materials, and the module is not allowed to withstand rain, snow, or liquid substances and mechanical damage.

## 3.2 Error message and Trouble Shooting

Customer can observe the module’s state by the LED status of its front panel in order to troubleshoot and slove the problem.

Table 4-1: Error message and trouble shooting

| LED status | Module state | Root cause | Solving method |
|---|---|---|---|
| All LED off | None | AC input switch not closed.<br>Poor contact of the AC connector. | Close the AC switch.<br>Check the AC connector. |
| Green LED blink | None | Wait for configuration and instructions from the upper controller or system | Send the command via the upper controller or system |
| Yellow LED ON | Current limited | Grid voltage at the low range and power derating.<br>DC voltage at the low range, power derating.<br>High ambient temperature, power derating.<br>Abnormal air inlet fan or blocked air duct, power derating. | Confirm the grid voltage.<br>Confirm the DC voltage.<br>Check the ambient temperature.<br>Check air ducts and fans. |
| Green LED | Recoverable | OTP shutdown.<br>Abnormal or asymmetric power grid. | Check the ambient temperature, air ducts and fans. |



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| | | | |
|---|---|---|---|
| ON | Alarm | Grid voltage amplitude OVP / UVP.<br>Grid voltage frequency OFP / UFP.<br>Internal / external communication failure.<br>DC voltage OVP / UVP.<br>AC side OCP.<br>DC side OCP.<br>Bus voltage OVP / UVP. | Check whether the grid voltage amplitude is within specification.<br>Check whether the grid voltage frequency is within specification.<br>Check whether communication lines are reliably connected and that address is correct.<br>Check the DC voltage amplitude.<br>check the load. |
| Red LED ON | Unrecoverable Fault | The discharge circuit failure.<br>Damaged Oring diode.<br>Short circuit in the AC side.<br>Short circuit in the DC side.<br>Damaged AC-side pre-charge resistor.<br>Fan failure.<br>The power device failure. | Check the AC load.<br>Check the AC side connection.<br>Checking the DC load.<br>Checking the DC side connection.<br>Check if the related device or circuit is normal. |

## 3.3 Replace the module

When there is a module failure in the system, replace it as follows.

- Disconnect the AC input switch and DC input switch from the system cabinet and wait more than 1 minute to ensure that the module ports have been fully discharged.
- Loosen and remove the module front panel fixing screws and disconnect the rear panel AC and DC connectors.
- Hold the faulty module by the handle and pull it out of the cabinet slowly by dragging it with the other hand, taking care to prevent burns when pulling it out.
- Push the new module into the cabinet slowly until it is fixed in place.
- Ensure that the AC and DC connectors on the rear panel are reliably connected and tighten the screws on the front panel.
- Turn on the AC input switch and DC input switch of the system cabinet.
- Check whether the system monitoring module can recognize the newly replaced module and complete the communication configuration and parameter setting.
- Start the system operation and check whether the voltage, current, power and temperature parameters of the module are normal.



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# Abbreviation

| | |
|---|---|
| DC | Direct current |
| AC | Alternating current |
| OVP | Over voltage protection |
| UVP | Under voltage protection |
| UFP | Under frequency protection |
| OFP | Over frequency protection |
| OTP | Over temperature protection |
| EMC | Electro Magnetic Compatibility |
| SGLCC | Steel Galvanized Aluminium Cold Common |
| AWG | American Wire Gauge |
| IP | Ingress protection |
| Cfm | Cubic feet per minute |
| USB | Universal Serial Bus |
| CAN | Controller Area Network |



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